Aurora Stays Q&A

Retrieval-Augmented Generation (RAG) POC

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Executive Summary

- Built a source-grounded RAG assistant for Aurora Stays hotel-policy questions.
- Four policy PDFs are converted to structured Markdown, chunked, embedded, and indexed in FAISS.
- Selected baseline: k=5, MMR lambda=0.50, fetch_k=20, CrossEncoder reranking.
- Held-out result: 9/10 cases passed; composite score 0.941.
- Average metrics: Faithfulness 0.993 | Recall 0.977 | Precision 0.808 | Relevancy 0.939.
- Main remaining issue: the Accessible King benchmark question is underspecified; a focused clarification experiment raised precision to 0.961.

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Business Problem Overview and Solution Approach

Problem

- Staff handle about 1,200 guest queries per week; roughly 40% require multiple policy documents.
- Manual lookup takes 8–12 minutes and creates a 15–20% risk of incomplete or incorrect answers.
- Property-specific fees, room features, cancellation rules, and accessibility guidance are easy to mix up.

Solution approach

- Extract policy PDFs with Docling; preserve headings and tables as Markdown.
- Use token-aware chunking, domain embeddings, FAISS retrieval, MMR, and CrossEncoder reranking.
- Generate concise answers with GPT-4o-mini using a strict retrieved-context-only prompt.

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Data Understanding

- Knowledge base: four selectable-text PDF policy guides.
- Coverage: room/property information; booking and cancellation; refunds and additional charges; conduct and accessibility.
- Source structure includes headings, policy sections, Markdown tables, and a six-property matrix.
- Properties represented: ASH-ATL, ASH-BOS, ASH-CHI, ASH-DEN, ASH-MIA, and ASH-SEA.
- OCR is disabled because the supplied PDFs contain selectable text.
- Chunks are enriched with source, section, property-code, and topic metadata for focused retrieval.

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RAG - Chunking - Embedding Model

- Embedding model: HarishMaths/Hotel-Policy-Embedding.
- Tokenizer maximum length: 128 tokens.
- Embeddings run on CPU with normalized vectors and progress rendering disabled.
- The same model is used when building and loading the FAISS index.
- Domain-specific embeddings improve matching for hotel-policy terms such as late checkout, pet fees, and room features.

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RAG - Chunking - Chunker

- Chunker: RecursiveCharacterTextSplitter.from_huggingface_tokenizer.
- Chunk size: 512 tokens; overlap: 64 tokens.
- Split priority: paragraph, line, heading, sentence, word, then character boundaries.
- Atomic property rows are added for property-matrix entries so a single property and fee can be retrieved precisely.
- Metadata is added after splitting to preserve section, topic, and property identity.

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RAG - Chunking - Chunk Documents

- Total chunks created: 43, including atomic property rows.

Example focused chunk

- ## Property-specific matrix | Property: ASH-MIA
- ASH-MIA Brickell | Parking: Valet, $55 per night | Pets: $100 per stay; 2 pets max
- One-property rows reduce contamination from unrelated properties and broad policy sections.
- Each chunk carries source, section, property_codes, and topics metadata.

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RAG - Vector Database

- Vector database: FAISS with 43 normalized document vectors.
- Index is persisted for the backend under backend/faiss_index.
- Final retrieval setting: k=5 chunks.
- Candidate search uses MMR lambda=0.50 and fetch_k=20.
- CrossEncoder ms-marco-MiniLM-L-6-v2 reranks candidates; topic and property metadata provide additional boosts.

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RAG - Vector Database - Retrieval Function

- Query: What are the parking fees at the Atlanta and Miami hotels?
- Property routing identifies ASH-ATL and ASH-MIA; topic routing identifies parking.
- Candidate chunks are retrieved with MMR, then reranked for direct relevance.

Relevant evidence

- ASH-ATL Midtown: self parking, $28 per night.
- ASH-MIA Brickell: valet, $55 per night.
- The final context contains the two requested property rows instead of unrelated policy sections.

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RAG - Generation System - Load the LLM

- Generator: OpenAI-compatible ChatOpenAI wrapper.
- Model: gpt-4o-mini.
- temperature=0.0 | top_p=1.0 | max_tokens=512.
- API key and base URL are loaded from environment/configuration rather than hard-coded.
- Deterministic settings favor repeatable, concise policy answers for evaluation and operations.

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RAG - Generation System - Generation Prompt

- Prompt inputs: user query + retrieved policy information.
- Use only information explicitly supported by retrieved policy text.
- Include relevant conditions, exclusions, limits, exceptions, and coverage details.
- Focus on the question; omit unrelated content and do not invent facts.
- If evidence is insufficient, return: The available policy documents do not contain enough information to answer this question.
- Additional guards focus Advance Purchase, Accessible King, and late-checkout questions on their relevant evidence.

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RAG - Generation System - RAG Function

- Pipeline: query -> retrieve top-k chunks -> generate grounded answer -> return answer and evidence.
- Example query: What is the cancellation policy for a flexible reservation?
- Example answer: Flexible rates can normally be canceled by 6:00 PM local property time one day before arrival without a room charge. Later cancellation or no-show may be charged one night plus tax; the confirmation controls if it differs.
- The response endpoint can return the answer alone or the answer plus relevant chunks for verification.

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Evaluation - Load the Evaluation Data

- Development evaluation file: data/evaluation.csv.
- 10 records with question, expected answer, and ground-truth relevant context.
- Sample question: What time can I check in at ASH-CHI Loop, and when do I need to be out?
- Expected answer: Standard check-in is 3:00 PM and standard check-out is 12:00 PM local property time; early arrival and late departure depend on availability and approval.
- Multi-context records use <> separators so relevant policy sections can be evaluated together.

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Evaluation - Define the Evaluation Model

- Judge model: gpt-4o through the DeepEval OpenAIModel wrapper.
- The generator remains gpt-4o-mini with temperature=0.0, top_p=1.0, and max_tokens=512.
- The stronger judge model scores factual support, retrieval coverage, retrieval focus, and answer relevance.
- Evaluations run synchronously so each test case has a stable, auditable metric result.

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Evaluation - Define the Evaluation Metrics

- Faithfulness >= 0.85: every important answer claim is supported by retrieved context.
- Contextual Recall >= 0.80: retrieved context contains information needed for the expected answer.
- Contextual Precision >= 0.75: relevant context is prioritized over unrelated content.
- Answer Relevancy >= 0.75: answer directly addresses the user question.
- Composite = 0.35F + 0.25Recall + 0.20Precision + 0.20Relevancy.
- A case passes only when all four metric thresholds are met.

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Evaluation - Build and Run the Evaluation

- Retrieval tuning matrix selected chunk_size=512, overlap=64, MMR lambda=0.50, and fetch_k=20.
- Dev proxy coverage: 0.836; held-out test proxy coverage: 0.850.
- Full DeepEval is the acceptance test; proxy coverage is used only to narrow candidates.
- The evaluation creates LLMTestCase objects with input, expected output, actual output, retrieval context, and reference context.
- Cases are classified as AUTO ANSWER, ASSISTED REVIEW, or MANUAL LOOKUP from the composite score.
- Result: baseline k=5 is the selected configuration by highest composite score.

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Evaluation - Load the Test Data

- Held-out file: data/test.xlsx.
- 10 unseen questions covering cancellation, discounts, room occupancy, accessibility, parking, breakfast, rate changes, checkout, pets, and incidental holds.
- Sample question: How many people can stay in a Double Queen room?
- Expected answer: Up to 4 guests, subject to fire-code and property occupancy limits; recheck the room and rate if the group size changes.
- The held-out set is used only for final comparison and selection of the baseline.

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Evaluation - Build and Run the Evaluation

Held-out DeepEval comparison (10 cases)

- Baseline k=5 F=.993 R=.977 P=.808 AR=.939 Composite=.941 Pass=9/10
- Hybrid k=5 F=.995 R=.979 P=.787 AR=.922 Composite=.935 Pass=8/10
- Hybrid k=6 F=.993 R=.982 P=.781 AR=.929 Composite=.935 Pass=8/10
- **Selected: baseline k=5 because it has the highest weighted composite score.**
- **Only remaining failure: Accessible King room inclusions; Contextual Precision=.451.**
- The benchmark question is underspecified. A focused clarification plus single-chunk retrieval scored Precision=.961 and passed all four metrics.

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Deployment - Backend - Serialize the Vector Database

- Backend artifact: backend/faiss_index/index.faiss.
- Metadata and document store: backend/faiss_index/index.pkl.
- The persisted FAISS index contains 43 normalized vectors from the policy knowledge base.
- The backend loads the trusted local artifact at startup; the Docker image copies the index into /app/faiss_index.
- The artifact must be rebuilt whenever source documents, chunking, or embeddings change.

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Deployment - Backend - Define the Flask Application

- Service entry point: backend/app.py; production command uses Gunicorn and app:rag_api.
- GET / returns a lightweight readiness message.
- POST /v1/relevant_chunks returns the top-k retrieved policy chunks.
- POST /v1/answer_with_relevant_chunks returns the generated answer and evidence chunks.
- POST /v1/answer returns the generated answer only.
- Defaults: k=5, gpt-4o-mini, temperature=0.0, top_p=1.0, max_tokens=512.
- Container binding: 0.0.0.0:7860; local smoke tests use port 5000.

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Deployment - Backend - Define the Dependencies

- Pinned backend dependencies are declared in backend/requirements.txt.
- Flask 3.1.3 and Gunicorn 26.2.0 provide the API service runtime.
- LangChain HuggingFace, LangChain Community, and LangChain OpenAI provide retrieval and generation integration.
- sentence-transformers 6.1.0 and faiss-cpu 1.15.0 provide embeddings and vector search.
- The Docker build installs CPU-only PyTorch before installing the requirements file.
- The embedding model is cached during image build so runtime can use offline mode.

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Deployment - Backend - Define the Dockerfile

- Base image: python:3.13.15-slim.
- Copies the backend application, FAISS artifact, and requirements into /app.
- Installs CPU PyTorch and the pinned backend dependencies without pip cache.
- Downloads HarishMaths/Hotel-Policy-Embedding at build time and sets HF_HOME=/app/huggingface.
- TRANSFORMERS_OFFLINE=1 and HF_HUB_OFFLINE=1 prevent runtime model downloads.
- Starts one Gunicorn worker on port 7860 with a 120-second request timeout.

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Deployment - Frontend - Define the Streamlit App

- Frontend entry point: frontend/app.py; it provides the Aurora Stays question-and-answer interface.
- Sidebar controls: k from 1 to 10, temperature from 0.0 to 1.0, and max_tokens from 64 to 2048.
- Users can select Answer only or Answer + Relevant Chunks.
- The backend address is supplied through BACKEND_URL; local default is http://localhost:5000.
- The interface displays the answer, retrieved evidence, and a clear API connectivity error when the backend is unavailable.
- Validated locally with Streamlit on port 8501 and the Flask API on port 5000.

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Deployment - Frontend - Define the Dockerfile

- Base image: python:3.13.15-slim.
- Copies the Streamlit application and installs requests 2.34.2 and streamlit 1.64.0.
- Binds Streamlit to 0.0.0.0:8501 for Codespaces or container access.
- XSRF protection is disabled in the supplied deployment command to simplify external access.
- Set BACKEND_URL at runtime so the frontend can call the backend service or forwarded port.

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Deployment - Frontend - Push the Files to GitHub

- Repository: github.com/raovutukoor1/aurora-stay-qna.
- Latest published commit: 7fda05a — Publish latest Aurora Stays evaluation and presentation.
- Published groups include backend and frontend source, FAISS artifacts, the final executed notebook, evaluation CSVs, and the presentation pack.
- The supplied template and generated PPTX/PDF presentation files are included for handoff.
- Credentials remain excluded: config/config.json, .env, and supplied token/config files are not tracked.

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Deployment - Inference - Define the Forwarded URL

- In GitHub Codespaces, open the Ports panel and forward the Streamlit port 8501.
- The public URL is generated per Codespace session; it is not hard-coded in the repository.
- Set BACKEND_URL to the backend service URL or its forwarded port before starting Streamlit.
- Production smoke tests used the parking-fee query and the ASH-CHI Loop pet-charge query.
- API keys belong in Codespaces Secrets or a private local config file, never in the forwarded URL or source code.

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Deployment - Inference - Relevant Chunks

- Endpoint: POST {BACKEND_URL}/v1/relevant_chunks.
- Payload: {"query": "What are the parking fees at the Atlanta and Miami hotels?", "k": 5}.
- Response fields: query, k, and relevant_chunks.
- Expected evidence includes ASH-ATL Midtown self parking at $28 per night and ASH-MIA Brickell valet at $55 per night.
- This endpoint is useful for checking retrieval quality before evaluating the generated answer.

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Deployment - Inference - Answer with Relevant Chunks

- Endpoint: POST {BACKEND_URL}/v1/answer_with_relevant_chunks.
- Payload includes query, k=5, temperature=0.0, and max_tokens=512.
- Response fields include answer, relevant_chunks, model_name, and the generation parameters used.
- The parking example returns a grounded comparison of the Atlanta and Miami fees with supporting chunks.
- Use POST /v1/answer when only the final answer is required; use this endpoint for auditability.

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APPENDIX

Production checklist

- Create config/config.json locally or set Codespaces Secrets for OPENAI_API_KEY and OPENAI_BASE_URL.
- Start the backend first, confirm the readiness response, then start Streamlit on port 8501.
- Run a property-specific query and an evidence-display query before wider use.

Known limitation

- The Accessible King case has Contextual Precision=.451 because the benchmark question is underspecified for the auto-answer strategy.
- The focused clarification experiment reached Precision=.961; the benchmark question and retrieval baseline remain unchanged.

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Production Readiness and Handoff Notes

- Selected baseline: k=5 with chunk_size=512, overlap=64, MMR lambda=0.50, fetch_k=20, and CrossEncoder reranking.
- Held-out result: 9/10 cases passed; weighted composite score 0.941.
- The application is runnable through the Flask backend and Streamlit frontend, locally or in GitHub Codespaces.
- Keep the benchmark question unchanged for evaluation; treat the remaining precision issue as a question/auto-answer limitation.
- Next operational step: add monitoring for insufficient context and review the single known failure before broader rollout.

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